

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Credit System) Degree Examinations
May - June 2023

19EC8X85 - SPACE TECHNOLOGY AND APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	State and explain Kepler's second law with the help of a neat diagram. What is the consequence of this law?	10	L*2	1	1
b)	Define the following terms for earth-orbiting satellites. (i) Subsatellite path, (ii) Line of apsides (iii) Ascending node, (iv) Descending node and (v) Inclination	10	L2	1	1
2. a)	What is ionosphere? Explain how this ionosphere effect the propagation of satellite signal and how those effects can be reduced.	10	L2	2	1
b)	Derive an expression for received power by a satellite antenna.	10	L2	2	1
3. a)	Name the facilities employed in satellites and ground stations communication to keep the satellites operational through commands. Explain in detail its functions.	10	L2	3	1
b)	What is transponder? With the help of a block diagram explain satellite wideband receiver.	10	L2	3	1
Unit – II					
4. a)	Why are navigation, guidance, and control subsystem needed for launch vehicles? Explain its functionalities.	10	L2	3	1
b)	Explain, why throttling is needed for satellite launch vehicle.	10	L2	3	1
5. a)	Draw the block diagram of Master Antenna TV System and explain functionality of each block.	10	L2	3	1
b)	Explain the objectives and applications of the remote sensing satellites.	10	L2	4	1
6. a)	What is Satellite transfer orbit and Hohmann transfer orbit? Explain with a supporting diagram.	10	L2	4	1
b)	Explain the factors considered for the selection of launch vehicles.	10	L2	4	1
Unit – III					
7. a)	Explain spade communication system with help of its diagrammatic representation.	10	L2	5	1
b)	Describe the coverage area, footprint, position of satellite and services provided of the Asian cellular system and Thuraya.	10	L2	5	1
8. a)	Explain with the help of a neat diagram, channeling arrangement for Intelsat SCPC system.	10	L2	5	1
b)	Explain in detail, how demand assignment may be carried out in Demand-assigned FDMA.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
